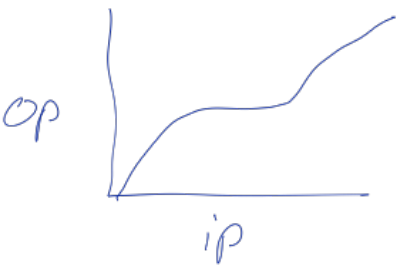
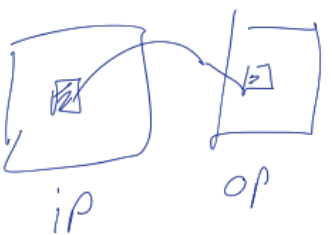
5/8/2025

L03 Spatial Filtering

In point operations



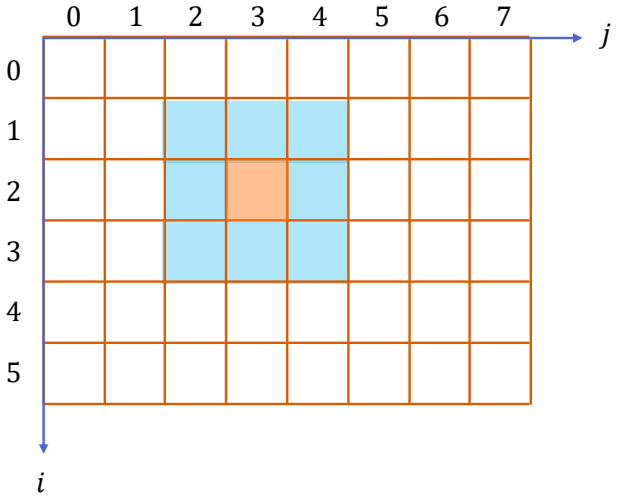


output point only depends on input point
and neighbours..

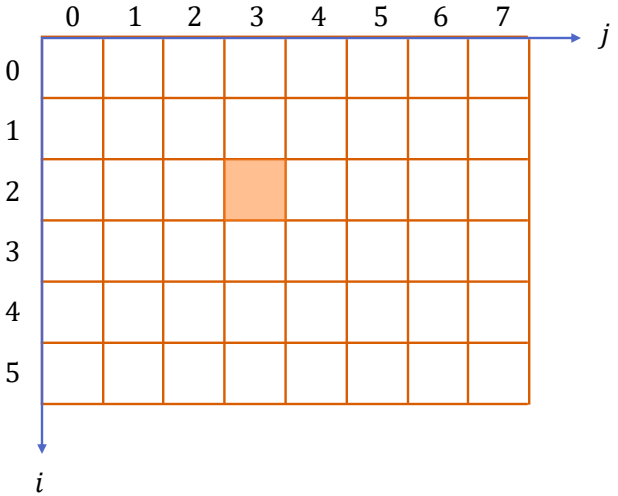
Introduction

- We use spatial filtering to enhance images, e.g., noise filtering.
- We can use the same technique to locate objects in images, called template matching. After completing this lesson, we would be able to process images using filters, as done in popular image processing software such as Photoshop.
- Intensity transformations affect the brightness (intensity level)of an image. The resulting value of a pixel after an intensity operation is just dependent on the original value of the same pixel.
- In contrast, the resulting value of a pixel after a spatial filtering operation depends on the neighborhood of the pixel in question. So, the space around the pixel in question matters, hence, the name spatial filtering.
- In linear spatial filtering we use the 2-D convolution operation.

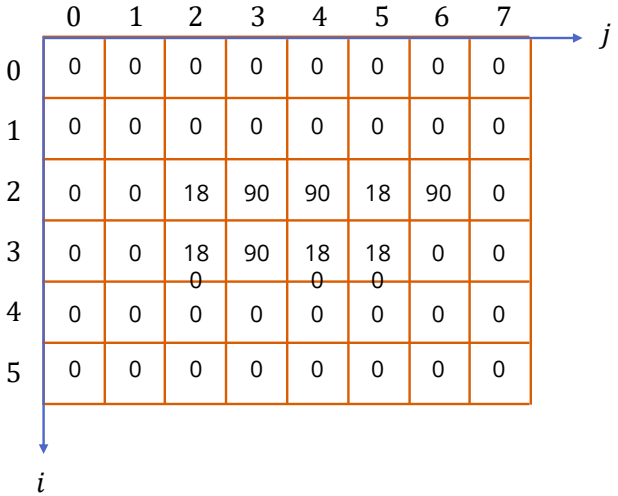
Spatial Filtering

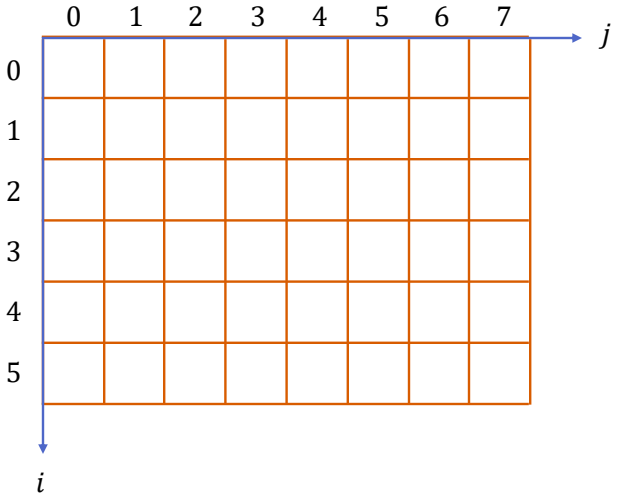


A 3 × 3 window (kernel or filter)



Spatial Filtering

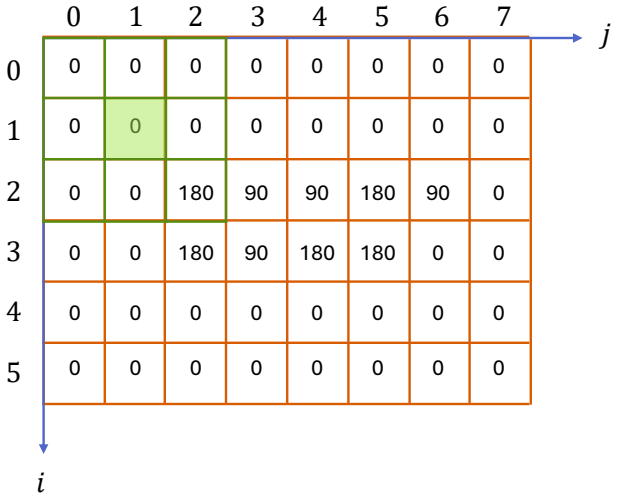


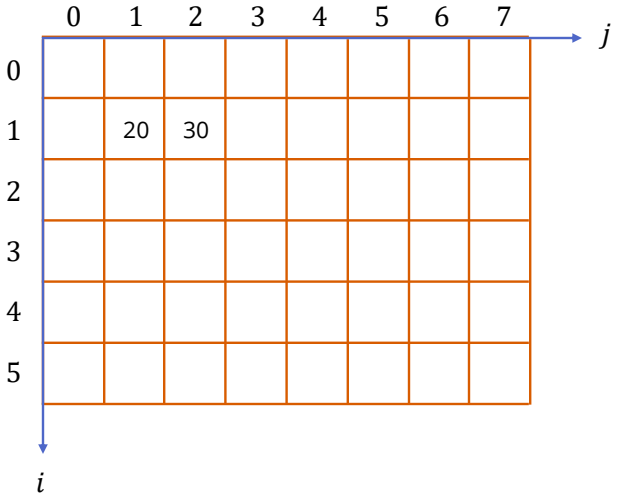


1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

kernel

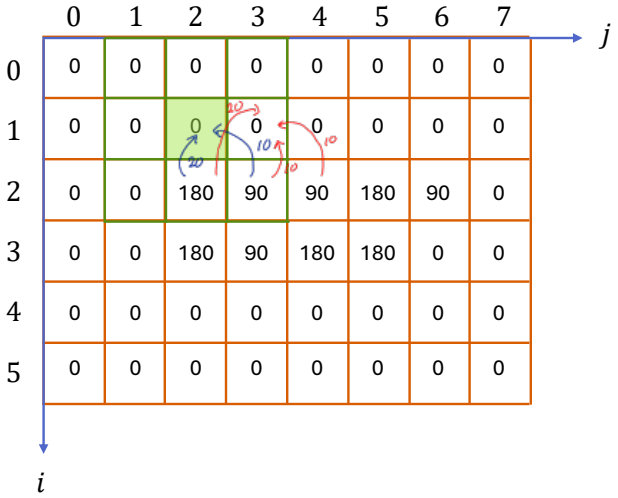
Spatial Filtering



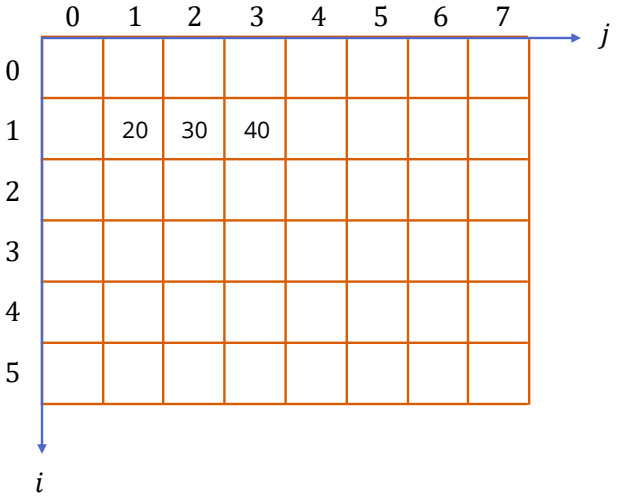


1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

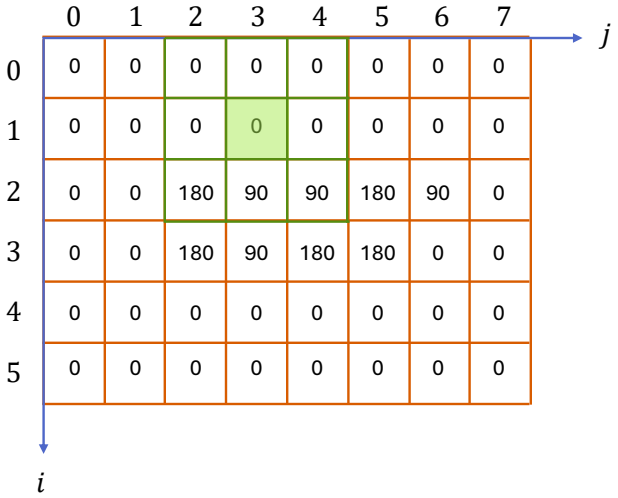
Spatial Filtering



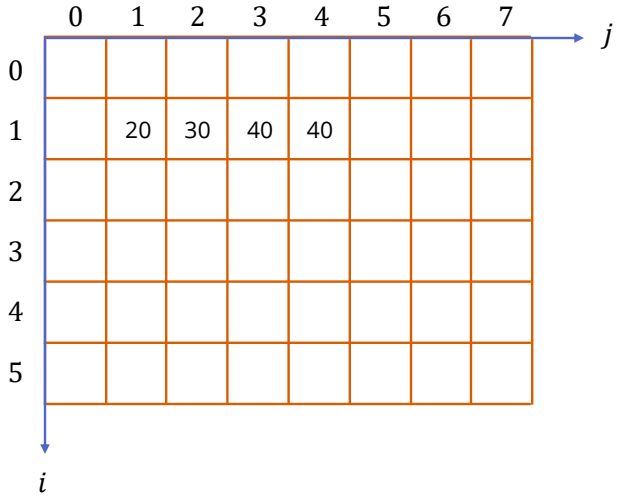
1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9



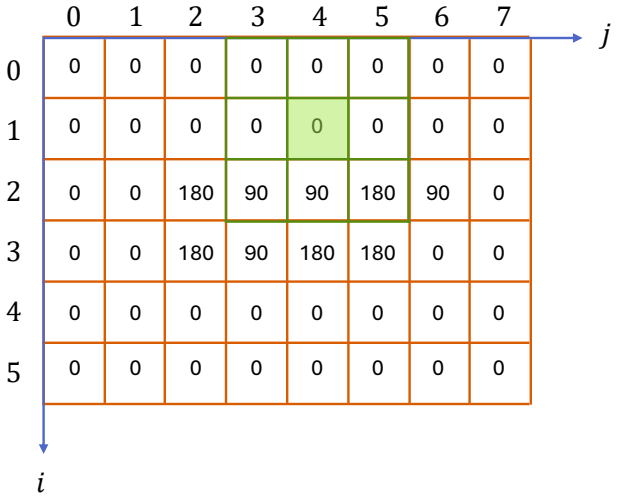
Spatial Filtering



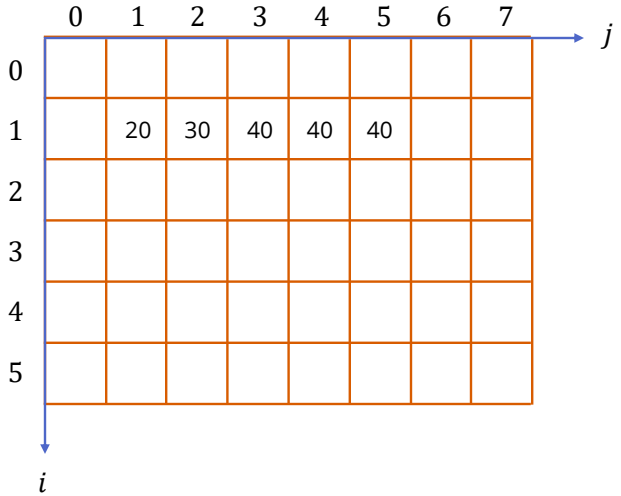
1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9



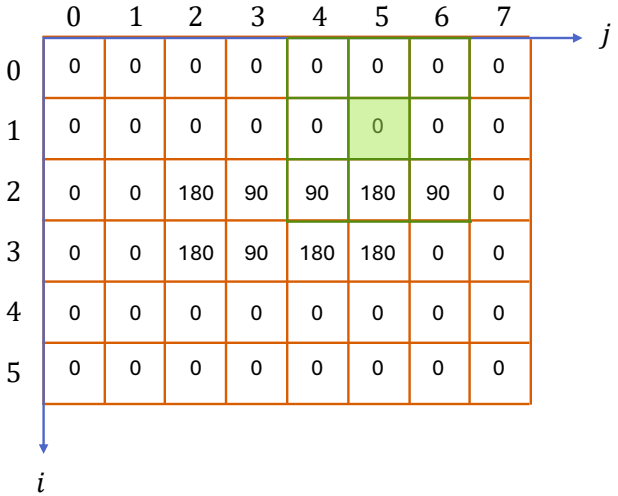
Spatial Filtering

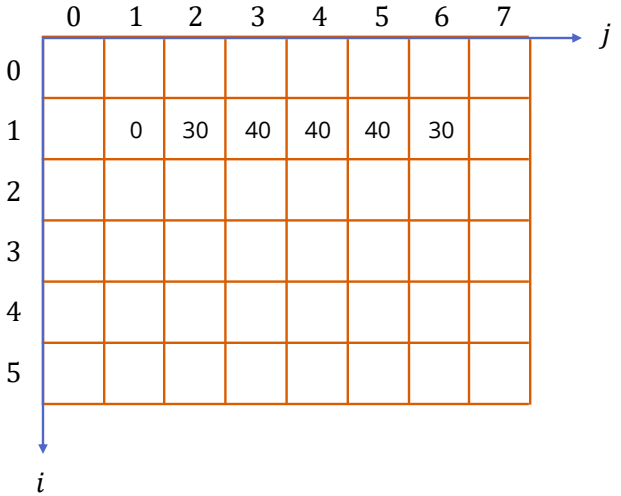


1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9



Spatial Filtering





1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

Spatial Filtering

i \ j	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0
2	0	0	180	90	90	180	90	0
3	0	0	180	90	180	180	0	0
4	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0	0

1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

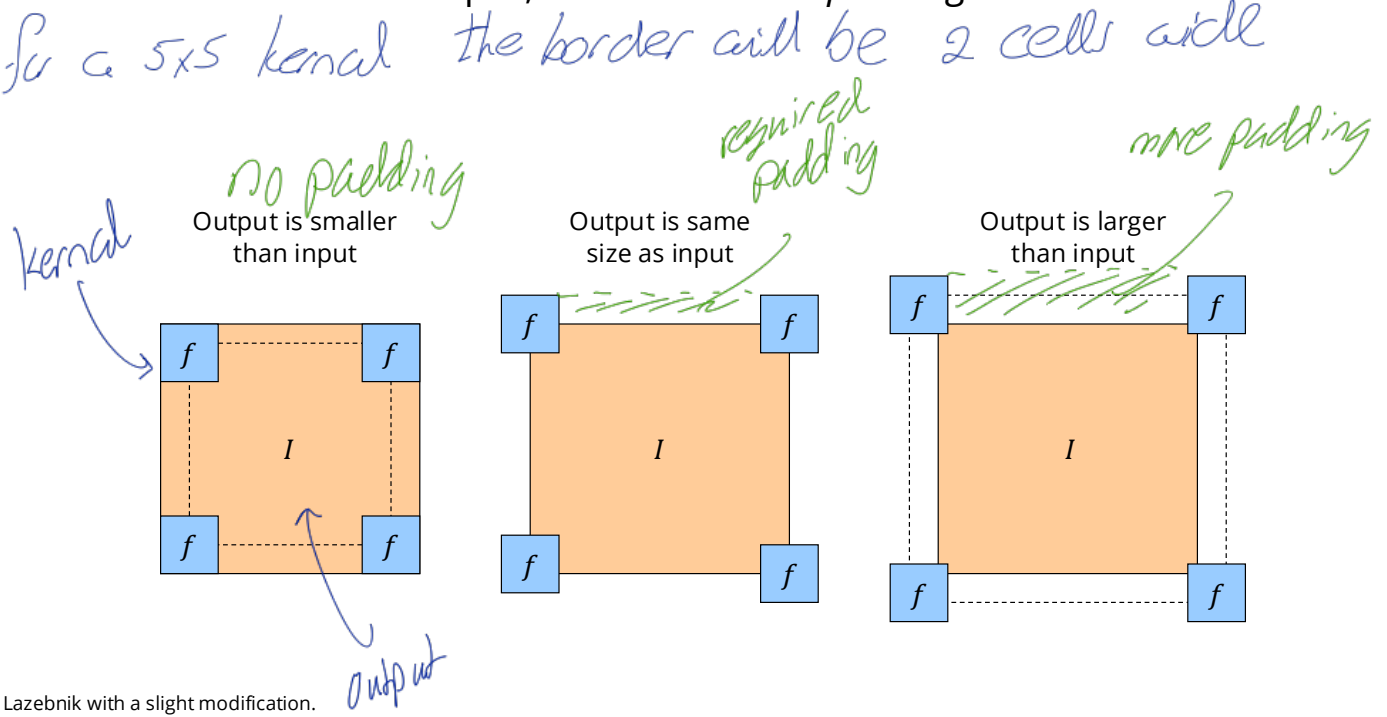
i \ j	0	1	2	3	4	5	6	7
0								
1		0	30	40	40	40	30	
2		40	60	90	90	80	50	
3		50	90	90	90	80	50	
4		20	30	50	50	40	20	
5								

when we try do fill the other cells the kernal will go outside

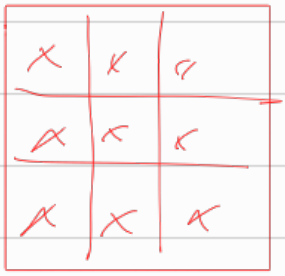
So we can use padding

Practical Details: Dealing with Image Boundaries

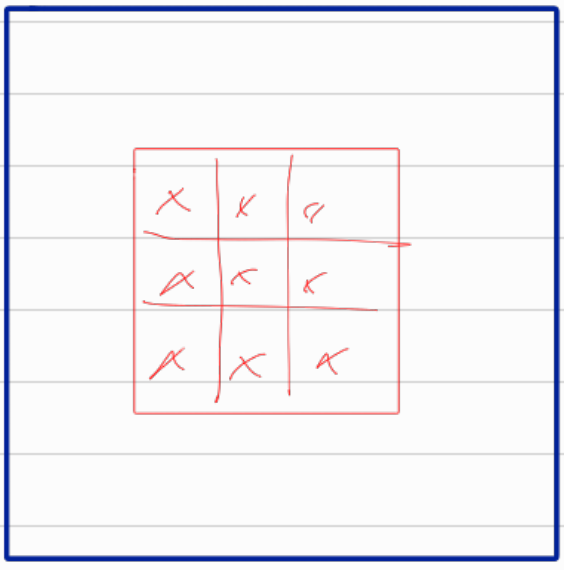
To control the size of the output, we need to use *padding*



Source: Lazebnik with a slight modification.



Kernal



we essentially multiply and add

This is the dot product

$$\underline{h} = \begin{bmatrix} x \\ x \\ x \\ \vdots \\ x \end{bmatrix} \quad \underline{f} = \begin{bmatrix} x \\ x \\ x \\ \vdots \\ x \end{bmatrix}$$

$$\underline{h} \cdot \underline{f} = |h||f| \cos\theta$$

max when $\cos\theta = 1$

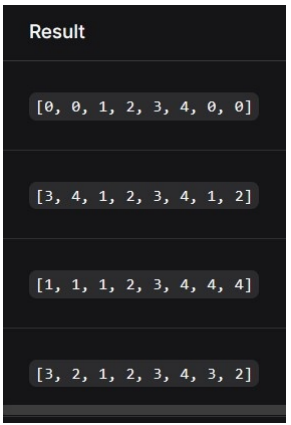
in other words the kernal matches the images.

So the output is highest when the image looks like the kernal

kernal looks for similar areas

Practical details: Dealing with Image Boundaries

- To control the size of the output, we need to use *padding*
- What values should we pad the image with?
  - Zero pad (or clip filter) BORDER_CONSTANT
  - Wrap around BORDER_WRAP
  - Copy edge BORDER_REPLICATE
  - Reflect across edge BORDER_REFLECT_101

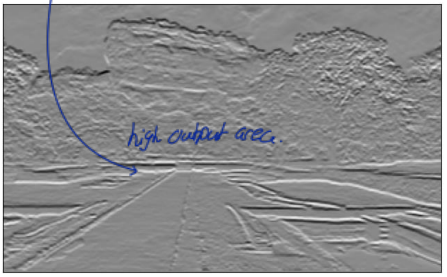


Source: S. Marschner

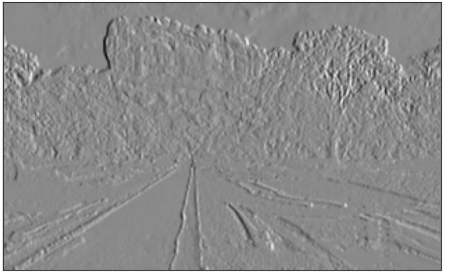
Examples of Effect of Kernel Choices

Original

Averaging



Sobel vertical



Sobel horizontal

1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

Averaging

-1	-2	-1
0	0	0
1	2	1

Sobel vertical

-1	0	1
-2	0	2
-1	0	1

Sobel horizontal

In sobel implementation we use correlation rather than convolution.

Averaging Using cv.filter2D

```
import cv2 as cv
import numpy as np
from matplotlib import pyplot as plt

im = cv.imread('images/sigiriya.jpg', cv.IMREAD_REDUCED_GRAYSCALE_2)
assert im is not None

kernel = np.ones((3,3),np.float32)/9
imavg = cv.filter2D(im, cv.CV_32F, kernel)

fig, axes  = plt.subplots(1,2, sharex='all', sharey='all', figsize=(18,9))
axes[0].imshow(im, cmap='gray')
axes[0].set_title('Original')
axes[0].set_xticks([]), axes[0].set_yticks([])
axes[1].imshow(imavg, cmap='gray')
axes[1].set_title('Averaging')
axes[1].set_xticks([]), axes[1].set_yticks([])
plt.show()
```

Load as grayscale and reduce size by half on both dimensions
use: quickly load a smaller grayscale version of an image

In filter2D it auto uses border reflect as padding and it does not flip the kernel

Sobel Filtering Using cv.filter2D

```
import cv2 as cv
import numpy as np
from matplotlib import pyplot as plt

im = cv.imread('images/sigiriya.jpg', cv.IMREAD_REDUCED_GRAYSCALE_2)
assert im is not None

sobel_x = np.array([[-1, -2, -1], [0, 0, 0], [1, 2, 1]])
sobel_y = np.array([[-1, 0, 1], [-2, 0, 2], [-1, 0, 1]])

im_x = cv.filter2D(im, cv.CV_64F, sobel_x)
im_y = cv.filter2D(im, cv.CV_64F, sobel_y)

fig, ax  = plt.subplots(1,2, sharex='all', sharey='all', figsize=(18,9))
ax[0].imshow(im_x, cmap='gray')
ax[0].set_title('Sobel X')
ax[0].set_xticks([]), ax[0].set_yticks([])
ax[1].imshow(im_y, cmap='gray')
ax[1].set_title('Sobel Y')
ax[1].set_xticks([]), ax[1].set_yticks([])
plt.show()
```

Central difference explained

In 1-D central difference

$$f'(x) = \frac{f(x+1) - f(x-1)}{2}$$

in sobel we use $[-1 \ 0 \ +1]$
it is similar
in the image we (+) the $x+1$ value and
(-) the $x-1$ value.

also in the other direction of the
sobel we have $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ which gives
a smoothing effect on the vertical axis.

even with just $[-1 \ 0 \ +1]$ we will
get the sobel effect but it will
be much more noisy.

---

for sobel we can use zero padding
- the math derivatives are 0
outside the boundary
- to prevent artificial gradients
- black borders mark unprocessed
areas.

4. When to Use Other Modes

Padding Mode	Ideal Use Case	Sobel Suitability
`constant` (zero)	Edge detection, technical imaging	★★★★★
`edge` (replicate)	Natural photo processing	★★★☆☆
`reflect`	Texture analysis, seamless processing	★★☆☆☆
`wrap`	Periodic signals (like FFT)	★☆☆☆☆

Convolution and Correlation

- Spatial filtering is, in fact, convolution.
- As the filters are typically symmetric—i.e., a 180◦ rotation results in the same kernel—correlation is equivalent to convolution.
- Correlation is also the scalar product between the kernel and the underlying image patch. Therefore, it is a measure of similarity between the kernel and the underlying image patch.
- As a result, when the kernel and the patch are “similar”, the output is high. In view of this, spatial filtering seeks for patches in the image that are similar to the kernel.
- Implementing filtering using loops (four nested for loops) in a non-C fashion is inefficient. Instead, use filter2D.

we did not flip the kernel. why.
convolution and correlation is similar for symmetric kernels

Convolution and Correlation

The convolution sum expression that we leaned in signal and systems is

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k].$$

In 2-D, as applicable in image processing, collation sum with a kernel $w[m, n]$ with non-zero values in $(m, n) \in ([-a, a], [-b, b])$ is

$$(w * f)[m, n] = w[m, n] * f[m, n] = \sum_{s=-a}^{a} \sum_{k=-b}^{b} w[s, t]f[m-s, n-t].$$

Correlation:

$$(w \circledast f)[m, n] = w[m, n] \circledast f[m, n] = \sum_{s=-a}^{a} \sum_{k=-b}^{b} w[s, t]f[m+s, n+t].$$

look similar

Example

Consider the image

$$f = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

and the filtering kernel

$$w = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

. Appropriately pad the image. Carry out a. correlation b. convolution.

after flipping or not, f remains the same.
add it to be 7×7

f =

0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0
0	0	0	0	0	0	0

c) (*)

8 8 7

6 5 4

3 2 1

Example

Consider the image

$$f_{\text{padded}} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \qquad w = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Correlation result and convolution result, respectively

$$(w \circledast f) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 9 & 8 & 7 & 0 \\ 0 & 6 & 5 & 4 & 0 \\ 0 & 3 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \qquad (w * f) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 4 & 5 & 6 & 0 \\ 0 & 7 & 8 & 9 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

if we do not flip, the result will be flipped

so we need to flip first to get it correct.

Convolution: Key Properties

1. Linearity: $\text{filter}(f_1 + f_2) = \text{filter}(f_1) + \text{filter}(f_2)$
2. Shift invariance: same behavior regardless of pixel location:
   $\text{filter}(\text{shift}(f)) = \text{shift}(\text{filter}(f))$
3. Theoretical result: any linear shift-invariant operator can be represented as a convolution.

Other Properties

1. Commutative: $a * b = b * a$
2. Conceptually no difference between filter and signal
3. Associative: $a * (b * c) = (a * b) * c$
4. Often apply several filters one after another: $(((a * b_1) * b_2) * b_3)$
5. This is equivalent to applying one filter: $a * (b_1 * b_2 * b_3)$
6. Distributes over addition: $a * (b + c) = (a * b) + (a * c)$
7. Scalars factor out: $ka * b = a * kb = k(a * b)$
8. Identity: unit impulse $e = [\ldots, 0, 0, 1, 0, 0, \ldots], a * e = a$

Suppose we have a two 3×3 kernels and a 1000×1000 image

Total computations $= 3^2 \times 1000^2 \times 2$

but due to associativity

$$\underset{3\times3}{a} * \left(\underset{3\times3}{b} * \underset{1000^2}{c} \right) = \underbrace{(a * b)} * c$$

$3^2 \times 3^2 + 3^2 \times 1000^2$

At the Edge

The filter window falls off the edge of the image.

Need to extrapolate the image.

Methods: various border types, image boundaries are denoted with '|'

- BORDER_REPLICATE: aaaaaa|abcdefgh|hhhhhhh
- BORDER_REFLECT: fedcba|abcdefgh|hgfedcb
- BORDER_REFLECT_101: gfedcb|abcdefgh|gfedcba
- BORDER_WRAP: cdefgh|abcdefgh|abcdefg
- BORDER_CONSTANT: iiiiii|abcdefgh|iiiiiii with some specified 'i'

Sharpening

(low freq)

high frequencies

(a) Original – (b) Smoothed = (c) Original – Smoothed

similar to high pass

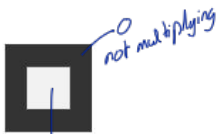
(d) Original + (e) Original – Smoothed = (f) Sharpened

smoothing is getting rid of high freq (blur)
sharpening is enhancing high freq
so we are adding the high freq content back to the original image.
containes edges

Box Filter vs. Gaussian Filter

- What's wrong with this filtering operation?
- What's the solution?

(a) Original



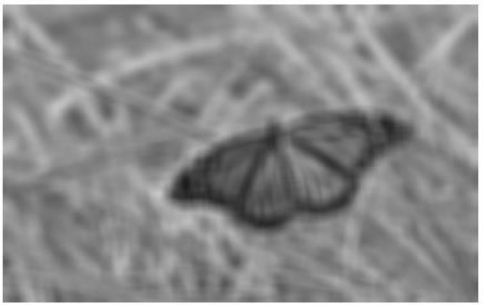
(b) Kernel (much zoomed)

(c) Box Filtered

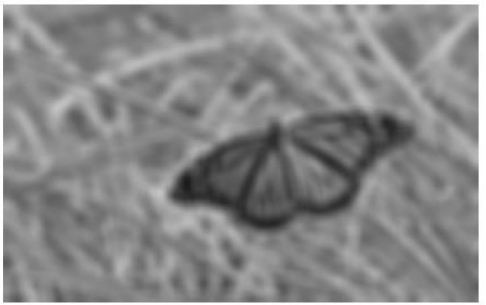
Gaussian has better spatial
- box has sharp cutoff and in freq domain it has sinc response which has strong side lobes (Gibbs phenomena)
- gaussian is spatially more / smooth weight falloff / center contribute more

Box Filter vs. Gaussian Filter

(a) Original



(b) Box Filtered



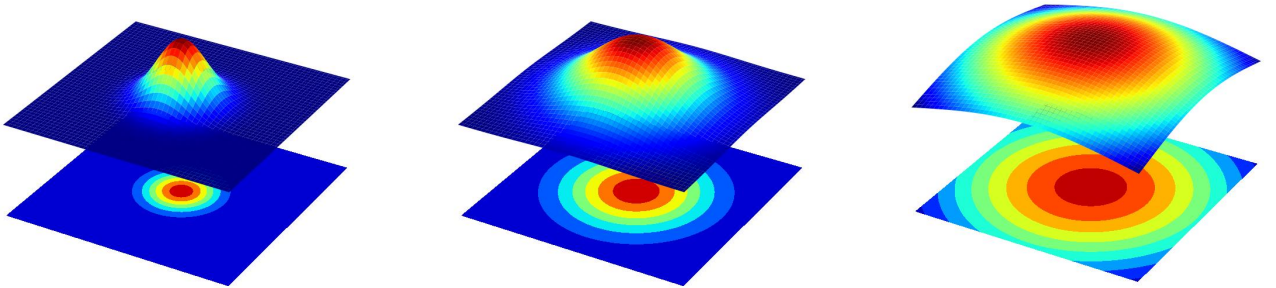
(c) Gaussian Filtered

Gaussian Kernel

$$G_\sigma(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

mean = 0

one std



- Constant factor at front makes volume sum to 1 (can be ignored when computing the filter values, as we should renormalize weights to sum to 1 in any case).
- The Gaussian function has infinite support, but discrete filters use finite kernels.

Compute a 2D 3×3 gaussian by hand. (un-normalized)

$$g(x,y) = e^{-\frac{(x^2+y^2)}{2\sigma^2}} \qquad \sigma = 1$$

$$= e^{-\frac{x^2+y^2}{2}}$$

compute for a 3×3 grid

(-1,1) $\frac{1}{e}$	(0,1) $\frac{1}{\sqrt{e}}$	$\frac{1}{e}$
$\frac{1}{\sqrt{e}}$	(0,0) 1	$\frac{1}{\sqrt{e}}$
$\frac{1}{e}$	$\frac{1}{\sqrt{e}}$	$\frac{1}{e}$

$\frac{1}{\sqrt{e}} = 0.607$

$\frac{1}{e} = 0.368$

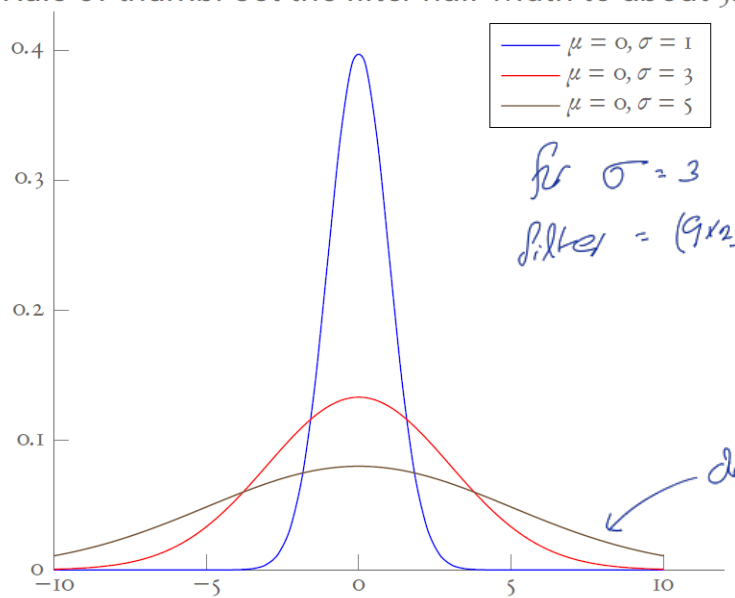
Now we have to normalize it by dividing by the sum of the kernel.

Otherwise an all white area can overflow.

Choosing Gaussian Kernel Width

$$g(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Rule of thumb: set the filter half-width to about 3σ.



for $\sigma = 3 \Rightarrow 3\sigma = 9$

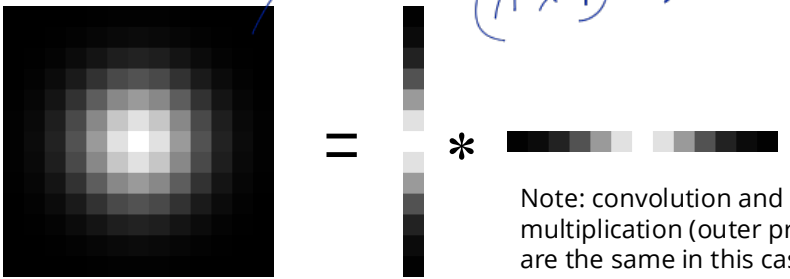
filter = $(9 \times 2) + 1 = 19$

doesn't go to 0

low blur

Separability of the Gaussian Filter

$$G(x,y) = \frac{1}{2\pi\sigma^2}\exp\left(-\frac{x^2+y^2}{2\sigma^2}\right) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{x^2}{2\sigma^2}\right)\ \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{y^2}{2\sigma^2}\right)$$



Note: convolution and matrix multiplication (outer product) are the same in this case

- The 2-D Gaussian can be expressed as the product of two functions, one a function of x and the other a function of y.
- In this case, the two Gaussians are the (identical) 1-D Gaussians.
- Separability means that a 2-D convolution can be reduced to two 1-D convolutions (one among rows and one among columns).
- What is the complexity of filtering an $n \times n$ image with an $m \times m$ kernel?

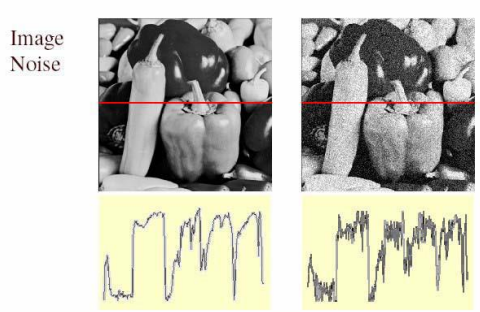
  $O(n^2m^2)$. m^2n^2
- What is the complexity if the kernel is separable?

  $O(n^2m)$ $mn^2 + mn^2$

image / kernel

Gaussian Noise

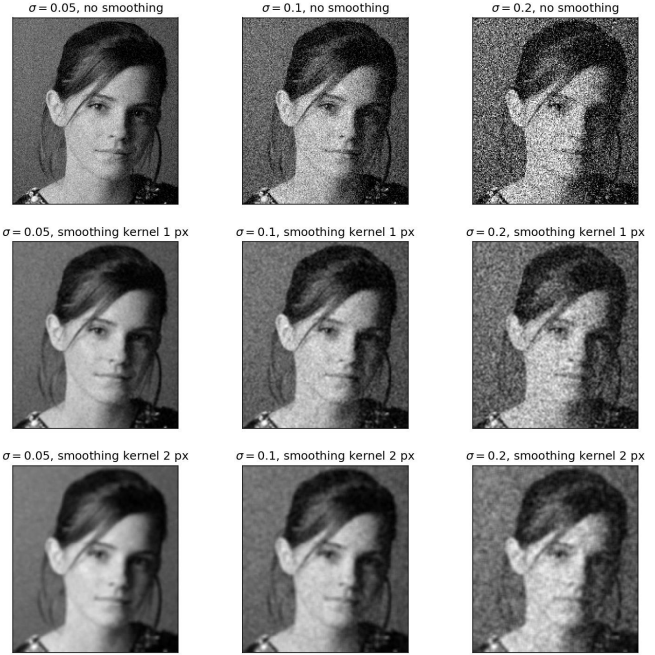
- Mathematical model: sum of many independent factors
- Good for small standard deviations
- Assumption: independent, zero-mean noise



$$f(x,y) = \overbrace{f(x,y)}^{\text{Ideal Image}} + \overbrace{\eta(x,y)}^{\text{Noise process}}$$

Gaussian i.i.d. ("white") noise:
$\eta(x,y) \sim \mathcal{N}(\mu, \sigma)$

when σ increase, noise level increase. more dot.
when the smoothing kernal size increase it reduces noise effect but blurs the image more.

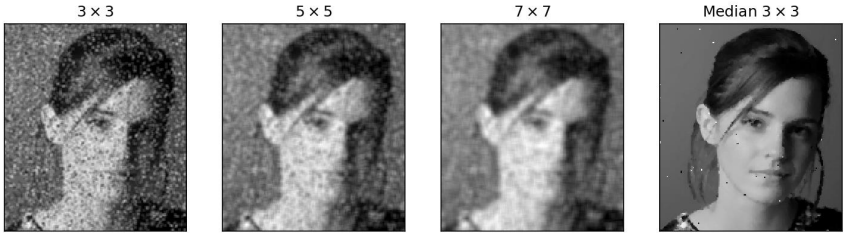


Reducing Gaussian Noise
Smoothing with larger standard deviations suppresses noise but also blurs the image.

Source: M. Hebert

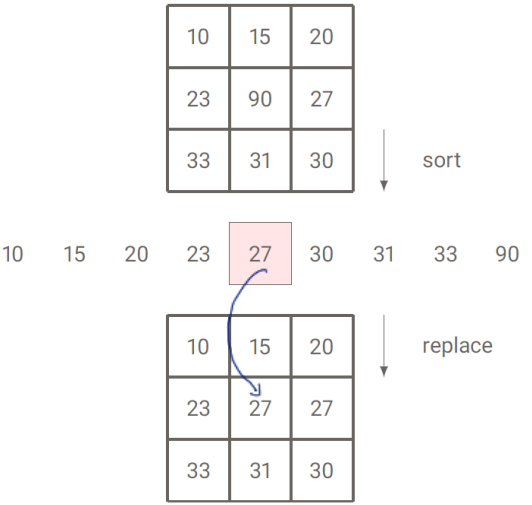
Reducing Salt-and-Pepper Noise: Median Filtering

not linear



Inability to reduce salt and pepper noise with Gaussian filtering

bigger window
⇒ remove more noise but more detail lost



A median filter operates over a window by selecting the median intensity in the window.
Is median filtering linear? NO

Source: K. Grauman

51	40	200			
50	150	300	60	70	80
45	55	65	75	85	95
			40	90	100

Bilateral Filter

Blur only a

What is the Bilateral Filter?

- A non-linear, edge-preserving, and noise-reducing smoothing filter.
- Combines domain (spatial) and range (intensity) filtering.
- Useful in denoising while preserving edges, unlike Gaussian smoothing.

Mathematical Formulation

what we did distance between pixels.

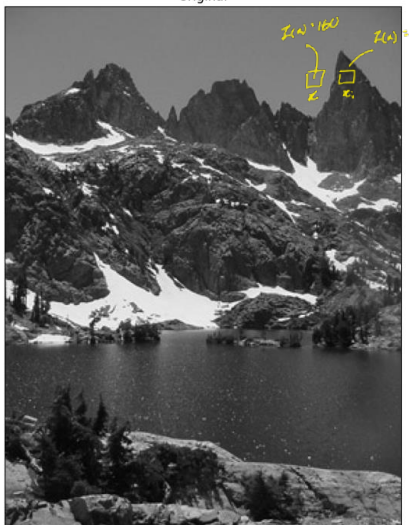
iso → $e^{-\infty} \approx 0$ no blur.

$$I_{\text{filtered}}(x) = \frac{1}{W_p} \sum_{x_i \in \Omega} G_{\sigma_s}(\|x_i - x\|) \cdot G_{\sigma_r}(\|I(x_i) - I(x)\|)$$

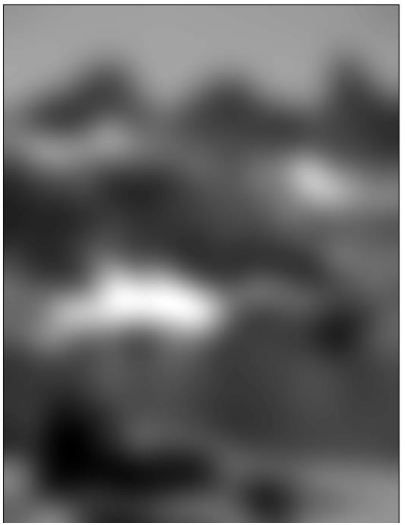
close say 1

completely blur

Original



Gaussian Filtered



Bilateral Filtered